

STATEMENT OF WORK

FOR

A26-013

Matrix Array Ultrasonic Testing NDI System

Prepared by

OBCA

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PREAMBLE

Commander's Safety Intent

The safety of everyone on Tinker AFB is our top priority. All Individuals must accept personal responsibility for safety and seek opportunities for improving the safety of our work environment. Everyone working on Tinker AFB is challenged to find, fix (if able) and report all hazards and near-misses. It is our shared duty to take every reasonable step and precaution to prevent accidents and preserve the health/safety of each Contractor and Government employee.

1 SCOPE. This Statement of Work (SOW) defines the effort required for the purchase of one (1) each Matrix Array Ultrasonic Testing NDI System and training at Tinker Air Force Base OK.

2 REQUIREMENTS

2.1 General Training shall be completed within three (3) business days of delivery. The Contractor shall provide to the Government Point Of Contact (GPOC) the date that Delivery, and Training. The Contractor shall be responsible for ensuring all services, parts and Original Equipment Manufacturer (OEM) components provided under this contract are in accordance with (IAW) the terms and conditions outlined herein, conform to the manufacturer's equipment specifications and this SOW.

2.2 Item Description and Requested Services: Tinker AFB requests the procurement of a Matrix Array Ultrasonic Testing NDI System with the following specifications:

Matrix Array Ultrasonic Testing (MAUT) NDI system including hardware and software SHALL meet the below technical requirements.

1. Technical Requirements Specification for a High-Resolution Volumetric Matrix Array UT System

1.0 General System Requirement

1.1 The proposed solution shall be a fully integrated, portable, and battery-operable matrix array ultrasonic testing (MAUT) system. The instrument, transducers, and software must be provided by a single manufacturer to ensure verified compatibility and seamless operation.

1.2.0 Phased Array Instrument Requirements

1.2.1 Channel Architecture: The instrument must possess a non-multiplexed, fully parallel hardware architecture capable of simultaneously driving and acquiring data from a minimum of 16,384 (128X128 matrix) individual transducer elements.

1.2.2 Real-Time Volumetric Data Processing: The instrument must be capable of performing real-time Full Matrix Capture (FMC) and subsequent Total Focusing Method (TFM) processing using the full aperture data from the specified 128x128 element transducers.

1.2.3 High-Resolution C-Scan Imaging: The system software must generate high-fidelity C-scan, B-scan, D-scan, and 3D volumetric images in real-time directly from the raw data acquired from the 128x128 element transducers. The C-scan image resolution must directly correspond to the physical element pitch of the specified transducers.

1.2.4 Data Throughput: The instrument must demonstrate a sustained data acquisition and transfer rate sufficient to handle the uncompressed volumetric data generated by a 16,384-element array without down-sampling or data reduction.

1.3.0 True 2D Matrix Array Transducer Requirements

1.3.1 Element Configuration: The system shall include two (2) True 2D Matrix Array transducers, each containing no fewer than 16,384 (128x128) individually addressable, monolithic piezoelectric elements arranged in a square grid configuration.

1.3.2 Frequency and Aperture: The transducer package must include the following:

- One (1) 5.0 MHz transducer with a 128x128 element matrix.
- One (1) 10.0 MHz transducer with a 128x128 element matrix.

1.4.0 System Integration and Performance

1.4.1 Verified System Operation: The supplier must provide documented proof that the proposed instrument (Section 2.0) has been specifically designed and optimized to operate the specified 128x128 element transducers (Section 3.0) as a complete, integrated system.

1.4.2 Acoustic Performance: The system must demonstrate the ability to generate fully focused acoustic images free of artifacts caused by multiplexing or element grouping. The point spread function of the system shall be consistent with the theoretical diffraction limit of the specified transducer apertures and frequencies.

1.5.0 Deliverables

1.5.1 The final deliverable must be a complete, field-ready kit including, at a minimum:

- One (1) MAUT Instrument meeting all requirements in Section 2.0.
- One (1) 5 MHz, 128x128 element True 2D Matrix Array Transducer.
- One (1) 10 MHz, 128x128 element True 2D Matrix Array Transducer.
- All necessary software licenses for data acquisition, advanced analysis (including FMC/TFM), and reporting.
- All required cables, power supply, and a rugged transport case.

- **Contractor shall provide Training on Tinker Air Force Base.**
- **The contractor must meet all the requirements outlined in the Statement of Work (SOW).**

2.3 Contractor Requirements

- The contractor shall deliver one (1) Matrix Array Ultrasonic Testing NDI System.
- The contractor shall provide Training

2.4 Training: The Contractor shall provide training on-site at Tinker Air Force Base for Tinker personnel. Training shall be for three days, for eight (8) hours per day and shall be specified in the Contractor's proposal

2.6 Hours of Operation: The Contractor shall perform the work required under this contract during the following days and hours: Monday through Friday, 0730 - 1600. If work hours are required outside the hours stated, the Contractor shall contact the Government POC for approval from the CO. Upon approval from the CO, the SME or GPOC escort Contractor personnel at all times.

2.6.1 These dates and hours exclude Federal Holidays, base closures due to inclement weather, and furlough days, as applicable.

3.0 On-Base Requirements

3.1 Environmental Protection & Safety: The Contractor shall be knowledgeable of and comply with all applicable federal, state and local laws, regulations, instructions and requirements regarding environmental protection and occupational safety and health. The Contractor shall comply with any laws, instructions or regulation changes during the term of this contract.

Contractors shall be compliant Tinker Air Force Base (TAFB) Supplement to Air Force Instruction (AFI) 32-7086, "Hazardous Materials Management", Tinker AFB procedures per **Tinker Air Force Base Hazardous Waste Management Plan (TAFB HWMP)**, and Complex procedures per Oklahoma City Air Logistics Complex (OC-ALC) Sup to AFI 32-7086, "Hazardous Material Management" and OC-ALCI 32-101 "Air Quality Record Keeping Procedure". Publications are available digitally on e-Publishing website at www.e-publishing.af.mil for downloading or ordering.

3.1.2 Hazardous Material (HAZMAT): If no HAZMAT will be brought on base or used in performance on this contract, the contractor shall certify "No HAZMAT will be brought on base" or "No HAZMAT Usage" by official correspondence. This certification shall be provided to the GPOC within ten (10) days after award. Upon COR receipt of the certification, the requirements of this section no longer apply. Appropriate penalties for the

contractor, or administrative and disciplinary action for the later shall be assessed if found in violation of above certification.

The Contractor shall not store chemicals or hazardous material (HAZMAT) on Tinker AFB. Contractors shall be compliant with HAZMAT management procedures per Oklahoma City Air Logistics Complex (OC-ALC) Supplement to AFI 32-7086, "Hazardous Materials Management". The Contractor shall maintain a log of all HAZMAT that is used on base. This usage log shall be provided to the GPOC monthly, the last business day of each month, or when the required work is completed. The GPOC will forward the records to the Unit Environmental Coordinator (UEC) no later than the fourth (4th) day of the following month or the first (1st) subsequent business day.

3.1.3 Safety Data Sheet (SDS) and Labeling: Upon COR receipt of the "No HAZMAT Usage" certification, the requirements of this section no longer apply.

The Contractor shall not transport, ship or use any chemical, material or substance used in the performance of this PWS on Tinker AFB before written approval is provided by 72 ABW/CEIE, Natural Infrastructure Management (Environmental Compliance). The Contractor shall submit to the GPOC, ten (10) calendar days after contract award date all Safety Data Sheets (SDSs) and a completed OC-ALC Form 152, SDS Information Sheet. The Contractor shall receive approval from the GPOC prior to shipping any chemical, material or substance to Tinker AFB. A copy of the manufacturer SDS shall accompany the shipment. Contractor shall comply with the effective dates for SDS requirements IAW the Occupational Safety and Health Administrations (OSHA) Hazardous Communication Standard, Title 29, Code of Federal Regulation (CFR) Part 1910.1200. The Contractor shall ensure containers of hazardous chemicals are labeled with the following information, prior to shipping to Tinker AFB IAW OSHA Hazardous Communication Standard, Title 29, CFR Part 1910.1200 (2012): product identifier; signal word; hazard statement(s); precautionary statement(s); pictogram(s); and name, address and telephone number of the chemical manufacturer, importer, or other responsible party. The GPOC will forward all SDS submittals to 72 ABW/CEIE for approval. A current manufacturer SDS shall be submitted to the GPOC if the Contractor changes manufacturer or vendor for any chemicals that were prior approved. The Contractor shall maintain copies of manufacturer SDSs for all chemicals and SDSs shall be readily accessible at all times by Contractor personnel as well as government employees.

3.1.4 Hazardous Waste: If no HAZWASTE will be generated during performance of this contract, the contractor shall certify "NO Hazardous Waste Generation" by official correspondence. This certification shall be provided to the GPOC within ten (10) days after award. Upon COR receipt of the "No Hazardous Waste Generation" certification, the requirements of this section no longer apply. Appropriate penalties for the contractor, or administrative and disciplinary action for the later shall be assessed if found in violation of above certification.

The Contractor shall dispose of contaminated solid waste, spent or processed chemical

solutions, used rags and hazardous waste in government furnished waste containers located in the shop or area the work is being performed. The Contractor shall notify the GPOC if the required waste containers are not available on site.

3.1.5 Discharges: Any discharges into Tinker's industrial waste treatment plant (IWTP) or sanitary sewer shall be approved and done IAW Tinker Industrial Wastewater Discharge Permit as instructed by 72 ABW/CEIE, Natural Infrastructure Management (Environmental Compliance). Notice of intent documentation shall be provided to the GPOC within ten (10) days after award. The GPOC will submit the required notice of intent documentation to the UEC for 72 ABW/CEIE approval. The Contractor shall not discharge into IWTP and sanitary sewer prior to approval issued by 72 ABW/CEIE. The Contractor shall maintain records of written approval issued by 72 ABW/CEIE for discharges into IWTP and sanitary sewer.

3.1.6 Off-Site Discharges: For discharges outside of Tinker AFB, the Contractor shall ensure wastewater discharges, that may contain pollutants resulting from any contracted service, go to a state or federal permitted treatment facility such as a municipal Publicly Owned Treatment Works (POTW). The Contractor shall provide official correspondence ten (10) calendar days after contract award date to the GPOC from the POTW that states the POTW is permitted for all pollutants that the Contractor may discharge during the contracted service.

3.1.7 Spills: Upon occurrence of any spill that could enter the storm system or cause any harmful environment effects, the Contractor shall immediately call 911 to report the incident. When possible, the Contractor shall use a base telephone to receive help from Tinker AFB 911. If using a cell phone, the Contractor shall request Tinker AFB 911. The Contractor shall immediately report all environmental violations to the GPOC and UEC. The Contractor shall provide five (5) copies of a typed or written report about the incident within four (4) clock hours as follows:

One (1) copy to CO, two (2) copies to the GPOC, a copy for UEC and one (1) copy for record. The format of the report shall be left up to the Contractor, but the report shall contain the following information:

- 1.) Name and contact number of person reporting the spill
- 2.) Date and time of incident
- 3.) Location and source of spill
- 4.) Substance or pollutant spilled
- 5.) Amount spilled and rate of discharge
- 6.) Any damages or injuries involved
- 7.) Extent of area impacted
- 8.) Potential hazards
- 9.) Actions taken
- 10.) Organizations contacted
- 11.) Remarks Section - that addresses the cause of the spill (e.g. human error, equipment, etc.) and whether 911 was called and which agencies responded.

3.1.8 Environmental Training: Upon COR receipt of the “No Hazardous Waste Generation” certification, the requirements of this section no longer apply.

Contractor personnel performing duties in the service of this PWS shall have training that acquaints the person with the details of Resource Conservation Recovery Act (RCRA) regulations in 40 CFR 260 through 279; applicable State Laws and regulations; DoD, Air Force and local requirements. This training can be obtained through creditable training institutions or companies. All training shall be approved by 72 ABW/CEIE, Natural Infrastructure Management (Environmental Compliance). RCRA training is available at <http://www.natlenvtrainers.com/RCRA-hazardous-waste.htm>; http://www.skillsoft.com/catalog/detail.asp?CourseCode=esh_sah_a34_sh_enus000000 and <http://www.lion.com/Storing-Hazardous-Waste-for-LQs> at the Contractor’s expense. Prior to obtaining training, the Contractor shall provide the name of the training institution or company and course description to the GPOC for approval within ten (10) calendar days after contract award date. Once written approval is received from the GPOC, the Contractor shall provide the environmental training certificates for employees performing duties in the service of this PWS to the GPOC, within thirty (30) calendar days after contract award date. In addition to the required environmental training, Contractors shall be compliant with Tinker AFB procedures per TAFB HWMP, “Hazardous Waste Management”.

3.2 Reserved

3.3 Housekeeping: The Contractor shall keep the work areas clean and neat IAW industry standards, OSHA, fire and safety standards, and this SOW.

3.4 Foreign Object Damage (FOD) and Dropped Object Prevention: Contractor personnel who work in or travel through areas near operational and production aircraft, engines, munitions, missiles, drones, space systems, support equipment, Aerospace Ground Equipment (AGE), trainers, and the Contractor personnel operating vehicles and equipment on flight lines, runways, taxiways, parking ramps, and in aircraft hangers and maintenance areas are required to receive the FOD/DOP training and to comply with AFI 36-2650, AFMC Supplement 1, paragraph 9.2., and A4.4.4. (Initial)/A4.4.5. (Refresher). The Contractor and Contractor personnel shall acknowledge the FOD/DOP training either by viewing the Digital Versatile Disc (DVD) provided by the GPOC or the You Tube site, <http://www.youtube.com/watch?v=tYcbodjFPnM>.

3.4.1 The Contract Manager shall provide written certification to the GPOC for each employee within ten (10) business days after the contract award date. The certification shall have the title of the training, the name of the person who received the training, and the date of the training. The Contractor shall also ensure all new employees view the FOD/DOP training and provide written certification of training to the GPOC prior to the employees starting work in the OC-ALC shops. Every 36 Months, the Contractor and Contractor personnel shall be required to review the FOD/DOP training and to provide written certification to the GPOC.

3.5 Tool Control and Accountability: While performing contract services in or around

Maintenance Production Shops, which includes ramp areas, the Contractor shall comply with AFI 21-101_AFCM Supplement 1, 21-101, Chapter 10 (all paragraphs) and OC-ALCI 21-112, paragraph 14.1. and 14.5.3. The Contractor shall provide the completed Tool Sheet Accountability prior to performing work in the OC-ALC industrial areas. The Tool Sheet Accountability is available from the GPOC. The Contractor or Contractor's personnel shall inventory the container or bag of tools at the beginning of each workday and at the end of each workday. The Contractor personnel shall immediately notify the GPOC of any tools that are missing or lost from the work area or container (e.g. tool or job box or tool bag). The GPOC will report any missing Contractor's tools which have not been located within one (1) hour after notification to the appropriate Government personnel. The CM shall ensure each employee on the job site receives a briefing from the GPOC about Housekeeping, FOD/DOP, and Tool Control prior to starting work in OC-ALC Industrial Areas

3.6 Privately Owned Motor Vehicles (POV): Contractor personnel shall comply with the directives pertaining to operation of POVs on Tinker AFB per AFMAN 31-116, in its entirety, "*Motor Vehicle Traffic Supervision*". The Contractor shall comply with Tinker AFB 24-302, in its entirety for the operation of POVs within Tinker AFB.

3.7 Visitors Pass: The Contractor shall contact the Government POC seven (7) business days before arriving on Tinker AFB in order to make arrangements for Visitors Passes. The Government POC will assign a sponsor for the Contractor from within the organization. The Contractor and the sponsor shall arrange to meet at Pass and Registration, building 6611 at the Tinker Gate, which is South of I-40 off Air Depot Blvd. [REDACTED]

3.8 Foreign Nationals: The Contractor shall notify the Government POC before sending a Foreign National representative to perform services. Special rules apply for foreign nationals visiting a US Government facility.

3.9 Safety and Health on Government Installations: While performing work under this contract on a Government installation, the Contractor shall comply with federal, state and local regulations, and AFI 91-203, "Air Force Consolidated Occupational Safety Instruction", as established by this contract and Appendix C. The Contractor shall notify the CO, Government POC, and Government Contract Administrator (CA) within one (1) business day of any injuries that occur to contractor personnel while they are on the installation. The Contractor shall notify the CO, Government POC, and Government CA within one (1) business day of damage to Government property or equipment during the execution of the contract.

3.10 Combating Trafficking in Persons: IAW DFARS PGI 222.17, the GPOC shall pursue, as appropriate, the following methods of monitoring the Contractor's performance regarding trafficking in persons such that non-compliances with FAR clause 52.222-50 are brought to the immediate attention of the Contracting Officer:

a. Keep the lines of communication open with the Contractor. At the Post-Award conference,

remind the Contractor of his contractual responsibilities to notify the government if the Contractor receives notification of any alleged violations to this policy or if actions have been taken against the Contractor employees, subcontractor's or subcontractor employees pursuant to the clause.

- b. When appropriate, encourage Contractor to complete Human Trafficking Awareness Training.
- c. Encourage the Contractor to take steps to investigate and eliminate slavery and human trafficking in their supply chains and to publish information for consumer awareness.
- d. Periodically access the Department of State's Trafficking in Person (TIP) website for updates and to view the latest reports. <http://www.state.gov/g/tip>

3.11 Commanders Safety Intent: The safety of everyone on Tinker AFB is our top priority. All Individuals must accept personal responsibility for safety and seek opportunities for improving the safety of our work environment. Everyone working on Tinker AFB is challenged to find, fix (if able) and report all hazards and near-misses. It is our shared duty to take every reasonable step and precaution to prevent accidents and preserve the health/safety of each contractor and government employee.

3.12 Speak up for safety: Safety surety is an integral part of our overall goal of establishing a world class safety and health program. Our "Speak Out for Safety" initiative ensures a partnership with contractors and government employees to further enhance safety to ensure we are looking out for all our partners and empowers all employees to actively participate in each other's safety.

A safe workplace is everyone's business. As part of the contract team, military, civilian and contractor personnel are empowered to Speak Out for Safety (S.O.S.) when they observe an unsafe act.

Events that present imminent danger to personnel or catastrophic damage to property we empower our employees to intercede to ensure continued safe operations.

3.13 Communicating Hazards of Workplace (CHOW): Contractor's onsite representative communicates all hazards being introduced to the worksite by the Contractor. This is accomplished prior to commencement of work and refreshed daily during performance, IAW Contractor's Site Safety Plan.

3.14 Emergency Procedures: The Contractor personnel shall follow the direction of the Government personnel with regard to emergency procedures, such as fire, tornado, active shooter, bomb threats, or another emergency. It is the responsibility of the Contractor to provide guidance and to establish procedures for its employees to respond to an emergency. It is the Contractor's responsibility to account for their employees following an emergency evacuation. The Contractor personnel shall participate with the Government personnel in all fire and tornado drills at no additional cost to the Government.